Refeena Refath

Early-career AI/ML engineer · RAG · FastAPI · Gemini API · ChromaDB · Python · API integration

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PROFESSIONAL SUMMARY

Information Technology graduate (B.Tech, CGPA 8.0) working as a freelance AI/ML engineer at Cloudroits, where I develop Campus Assist: a retrieval-augmented AI assistant that answers university questions from approved documents, built with FastAPI, ChromaDB and the Gemini API and integrated with Odoo. Currently completing the Applied AI and Machine Learning Essentials programme from Vishlesan i-Hub, IIT Patna with Masai School. I enjoy connecting AI models to real processes safely.

SKILLS

Generative AI: RAG, LLM API integration (Gemini API), ChromaDB vector search, document-grounded assistants, prompt engineering
Backend and integration: FastAPI, REST APIs, JWT authentication, role-based access, Odoo integration, automated testing
Machine learning: Classification (Random Forest, SVM, Logistic Regression, XGBoost), model evaluation, NLP basics
Programming: Python, Java, JavaScript, SQL, C++, HTML, CSS
Data and tools: pandas, NumPy, matplotlib, scikit-learn, Google BigQuery, React, Git, GitHub, VS Code

EXPERIENCE

AI/ML Engineer (freelance contract) — Cloudroits Jul 2026 – Present

Campus Assist: AI-powered university knowledge assistant

Tools: Python, FastAPI, RAG, ChromaDB, Gemini API, Odoo, SQL, JWT authentication

- Developed Campus Assist, a document-grounded AI assistant that answers university questions from uploaded reference documents, using FastAPI, ChromaDB and the Gemini API (retrieval-augmented generation).
- Implemented document ingestion and retrieval: text chunking, vector search in ChromaDB, and answers linked to their source documents.
- Implemented role-based access control and institution-level data isolation so each university tenant sees only its own information.
- Added document lifecycle controls (validity dates, version supersession) to reduce the risk of answers coming from outdated policies.
- Integrated the AI backend with an Odoo university management system and OdooBot for in-app question answering.

PROJECTS

TrendPulse — trending-news data pipeline 2026

Collects trending Hacker News stories, groups them by topic and turns them into charts.

Tools: Python, requests, pandas, NumPy, matplotlib, Hacker News API | github.com/refinah20/trendpulse-refeena-refath-z

- Pulled top stories from the Hacker News public API with Python (requests) and sorted them into topics such as technology, world news, sports and science using keyword rules.
- Built a four-stage pipeline (collect, clean, analyse, visualise) with pandas and NumPy, and produced charts with matplotlib, such as the top 10 stories by score.

Intrusion Detection System — machine-learning model comparison 2024

Flags network connections that look anomalous, as an early warning of possible attacks.

Tools: Python, pandas, scikit-learn, XGBoost | github.com/Refinah/intrusiondetection

- Trained and compared Random Forest, SVM, Logistic Regression and XGBoost classifiers on about 25,000 labelled network-traffic records to label connections as normal or anomalous.

EDUCATION AND TRAINING

Applied Artificial Intelligence and Machine Learning Essentials May 2026 – Oct 2026

Vishlesan i-Hub, IIT Patna, in collaboration with Masai School · Online certificate programme. Expected completion: 25 October 2026.

B.Tech, Information Technology Completed Dec 2025

Saveetha School of Engineering, SIMATS, Chennai · CGPA: 8.0 / 10

Java Full Stack Development (training course) Feb 2024 – Sep 2024

Besant Technologies

CERTIFICATIONS

C++ Programming — CodSoft · WordPress — Coursera · Google BigQuery — Coursera